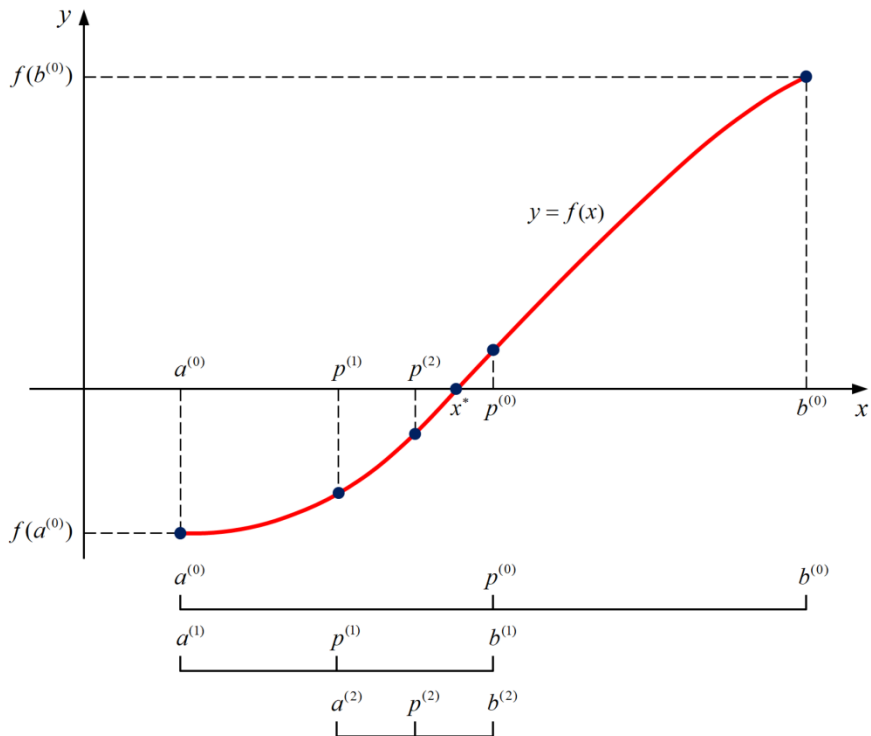


Bisection Method



Bracketing Methods: There is a root, x^* , between a and b .

$$f(x^*) = 0, a < x^* < b \rightarrow f(a) \cdot f(b) < 0$$

Bisection Method: $p = \frac{a + b}{2}$

$$f(a) \cdot f(p) \leq 0 \rightarrow a < x^* \leq p \rightarrow \begin{cases} a^{(k)} = a^{(k-1)} \\ b^{(k)} = p^{(k-1)} \end{cases}$$

$$f(a) \cdot f(p) > 0 \rightarrow p < x^* \leq b \rightarrow \begin{cases} a^{(k)} = p^{(k-1)} \\ b^{(k)} = b^{(k-1)} \end{cases}$$

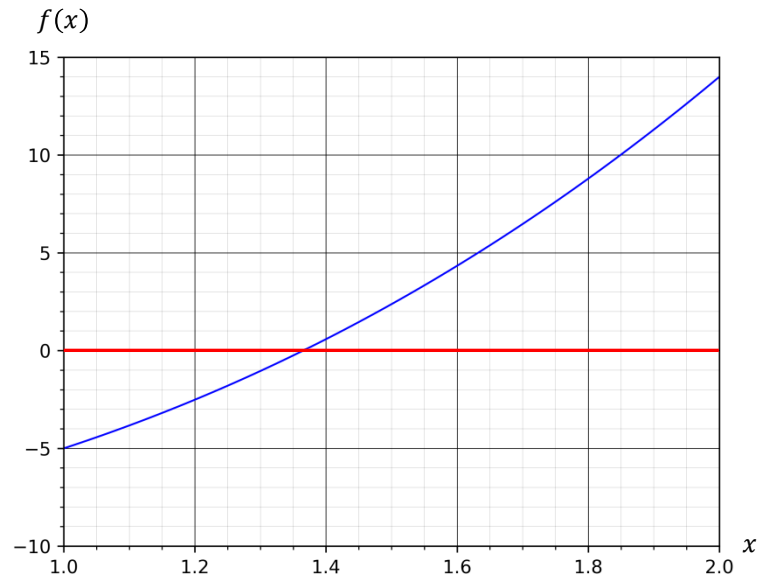
Relative Error: $\varepsilon^{(k)} = \frac{|p^{(k)} - p^{(k-1)}|}{|p^{(k)}|}$

Relative Residual: $\varepsilon^{(k)} = \frac{|r^{(k)}|}{|r^{(0)}|} = \frac{|f(p^{(k)})|}{|f(p^{(0)})|}$

Bisection Method

Example 02-01: Find a root of equation $x^3 + 4x^2 - 10 = 0$ in the interval $[1,2]$ using bisection method with $\varepsilon_t = 10^{-6}$ for relative error.

$$\begin{aligned} a = 1, f(a) &= -5 \\ b = 2, f(b) &= 14 \end{aligned} \quad \rightarrow \quad f(a) \cdot f(b) < 0$$



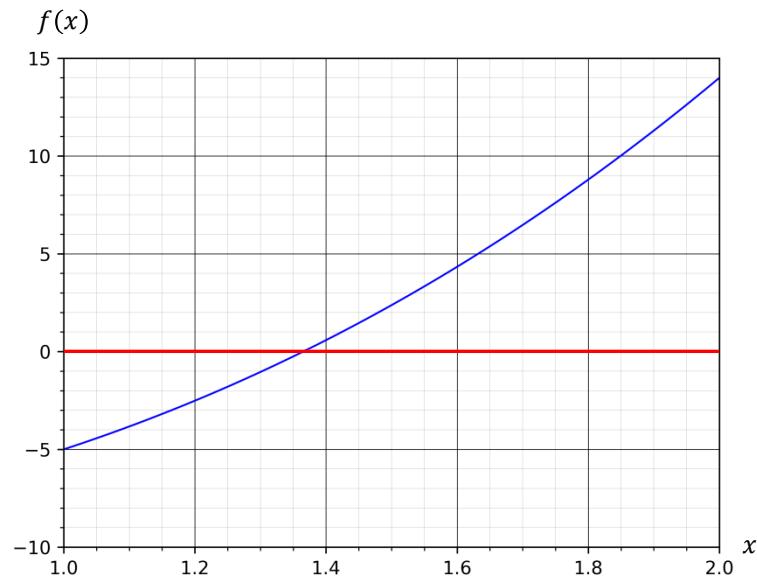
Bisection Method

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$$k = 0$$

$$a^{(0)} = 1, b^{(0)} = 2$$



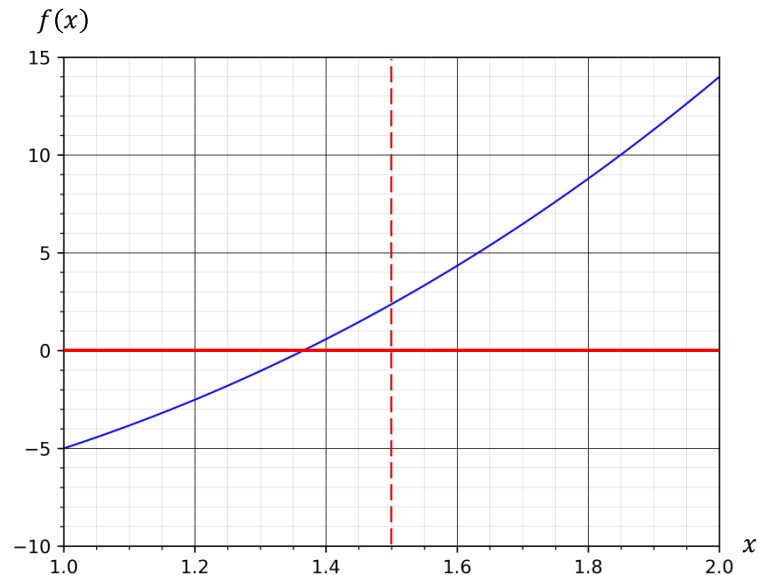
Bisection Method

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$$k = 0$$

$$a^{(0)} = 1, b^{(0)} = 2 \quad \rightarrow \quad p^{(0)} = \frac{1+2}{2} = 1.5$$



Bisection Method

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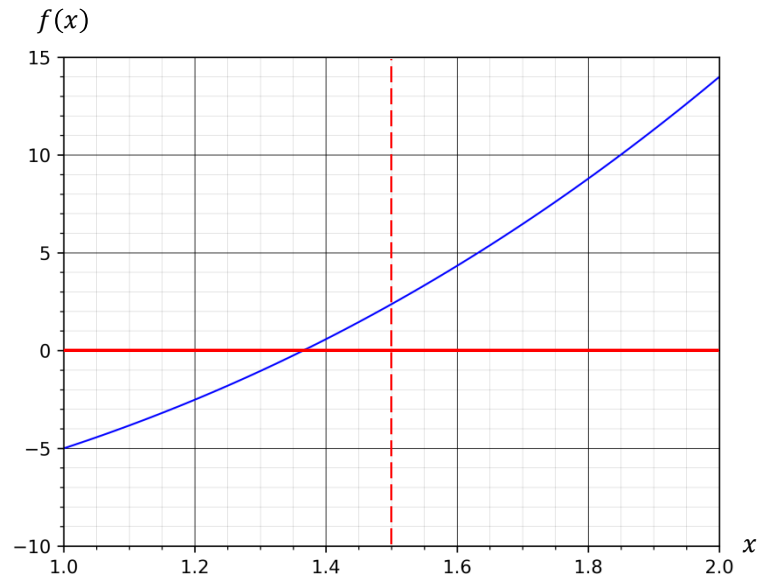
$$\begin{aligned} a = 1, f(a) &= -5 \\ b = 2, f(b) &= 14 \end{aligned} \quad \rightarrow \quad f(a) \cdot f(b) < 0$$

$$k = 0$$

$$a^{(0)} = 1, b^{(0)} = 2 \quad \rightarrow \quad p^{(0)} = \frac{1+2}{2} = 1.5$$

$$k = 1$$

$$f(a^{(0)}) = -5, f(p^{(0)}) = 2.375 \quad \rightarrow \quad f(a^{(0)}) \cdot f(p^{(0)}) < 0$$



Bisection Method

Example 02-01: Find a root of equation $x^3 + 4x^2 - 10 = 0$ in the interval $[1,2]$ using bisection method with $\varepsilon_t = 10^{-6}$ for relative error.

$$\begin{aligned} a = 1, f(a) &= -5 \\ b = 2, f(b) &= 14 \end{aligned} \quad \rightarrow \quad f(a) \cdot f(b) < 0$$

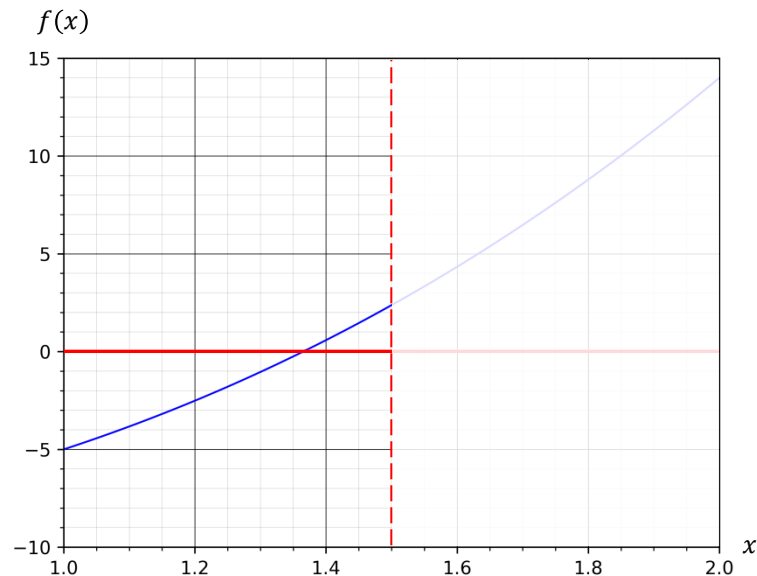
$k = 0$

$$a^{(0)} = 1, b^{(0)} = 2 \quad \rightarrow \quad p^{(0)} = \frac{1+2}{2} = 1.5$$

$k = 1$

$$f(a^{(0)}) = -5, f(p^{(0)}) = 2.375 \quad \rightarrow \quad f(a^{(0)}) \cdot f(p^{(0)}) < 0$$

$$a^{(1)} = a^{(0)} = 1, b^{(1)} = p^{(0)} = 1.5$$



Bisection Method

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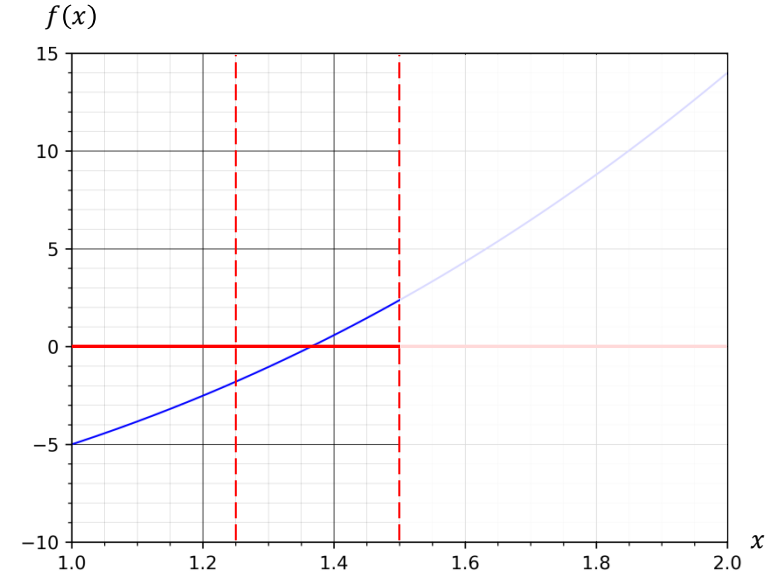
$k = 0$

$$a^{(0)} = 1, b^{(0)} = 2 \quad \rightarrow \quad p^{(0)} = \frac{1+2}{2} = 1.5$$

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$$f(a^{(0)}) = -5, f(p^{(0)}) = 2.375 \quad \rightarrow \quad f(a^{(0)}) \cdot f(p^{(0)}) < 0$$

$$a^{(1)} = a^{(0)} = 1, b^{(1)} = p^{(0)} = 1.5 \quad \rightarrow \quad p^{(1)} = \frac{1+1.5}{2} = 1.25$$



Bisection Method

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$k = 0$

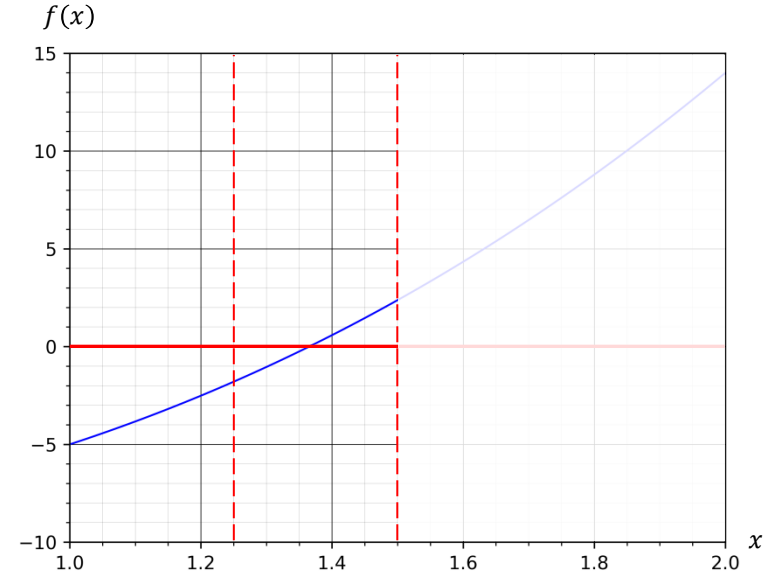
$$a^{(0)} = 1, b^{(0)} = 2 \quad \rightarrow \quad p^{(0)} = \frac{1+2}{2} = 1.5$$

$k = 1$

$$f(a^{(0)}) = -5, f(p^{(0)}) = 2.375 \quad \rightarrow \quad f(a^{(0)}) \cdot f(p^{(0)}) < 0$$

$$a^{(1)} = a^{(0)} = 1, b^{(1)} = p^{(0)} = 1.5 \quad \rightarrow \quad p^{(1)} = \frac{1+1.5}{2} = 1.25$$

$$\varepsilon^{(1)} = \frac{|p^{(1)} - p^{(0)}|}{|p^{(1)}|} = \frac{|1.25 - 1.5|}{|1.25|} = 0.2 > \varepsilon_t$$



Bisection Method

Example 02-01: Find a root of equation $x^3 + 4x^2 - 10 = 0$ in the interval $[1,2]$ using bisection method with $\varepsilon_t = 10^{-6}$ for relative error.

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$k = 0$

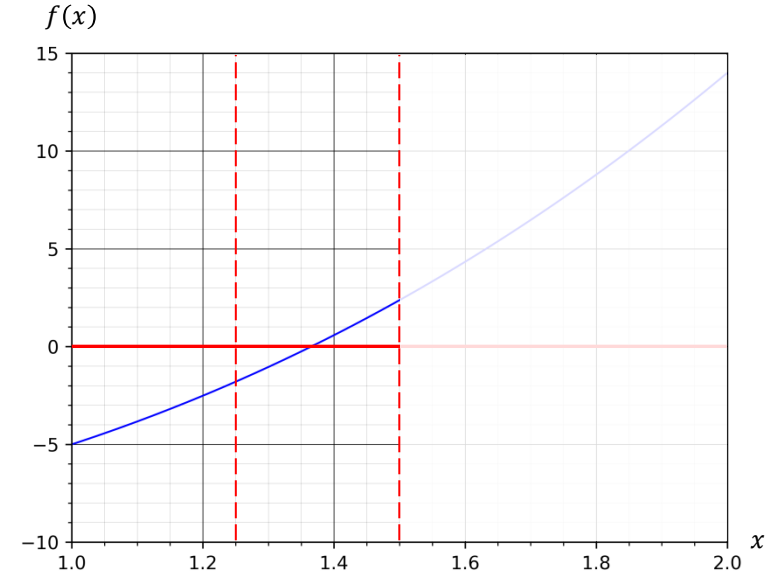
$$a^{(0)} = 1, b^{(0)} = 2 \quad \rightarrow \quad p^{(0)} = \frac{1+2}{2} = 1.5$$

$k = 1$

$$f(a^{(0)}) = -5, f(p^{(0)}) = 2.375 \quad \rightarrow \quad f(a^{(0)}) \cdot f(p^{(0)}) < 0$$

$$a^{(1)} = a^{(0)} = 1, b^{(1)} = p^{(0)} = 1.5 \quad \rightarrow \quad p^{(1)} = \frac{1+1.5}{2} = 1.25$$

$$\varepsilon^{(1)} = \frac{|p^{(1)} - p^{(0)}|}{|p^{(1)}|} = \frac{|1.25 - 1.5|}{|1.25|} = 0.2 > \varepsilon_t \quad \rightarrow \quad \text{continue}$$



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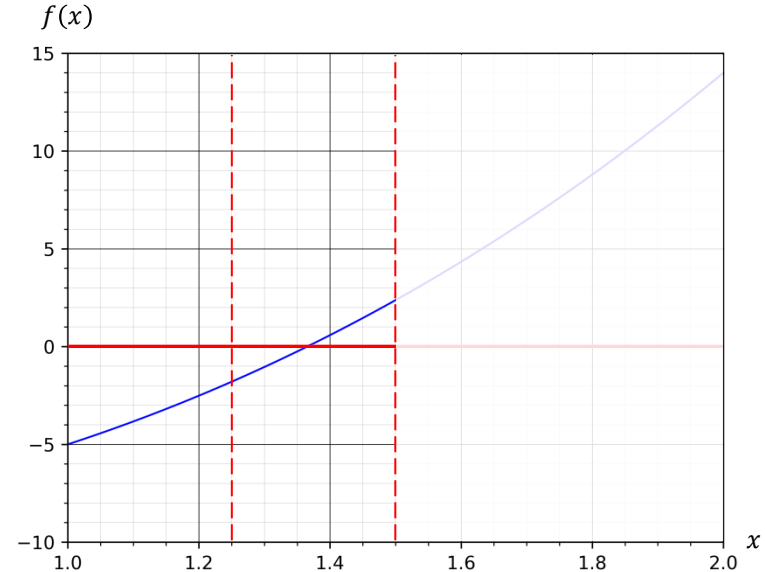
$$f(a^{(0)}) = -5, f(p^{(0)}) = 2.375 \quad \rightarrow \quad f(a^{(0)}) \cdot f(p^{(0)}) < 0$$

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$$\varepsilon^{(1)} = \frac{|p^{(1)} - p^{(0)}|}{|p^{(1)}|} = \frac{|1.25 - 1.5|}{|1.25|} = 0.2 > \varepsilon_t \quad \rightarrow \quad \text{continue}$$

$k = 2$

$$f(a^{(1)}) = -5, f(p^{(1)}) = -1.797 \quad \rightarrow \quad f(a^{(1)}) \cdot f(p^{(1)}) > 0$$



Bisection Method

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$k = 0$

$$a^{(0)} = 1, b^{(0)} = 2 \quad \rightarrow \quad p^{(0)} = \frac{1+2}{2} = 1.5$$

$k = 1$

$$f(a^{(0)}) = -5, f(p^{(0)}) = 2.375 \quad \rightarrow \quad f(a^{(0)}) \cdot f(p^{(0)}) < 0$$

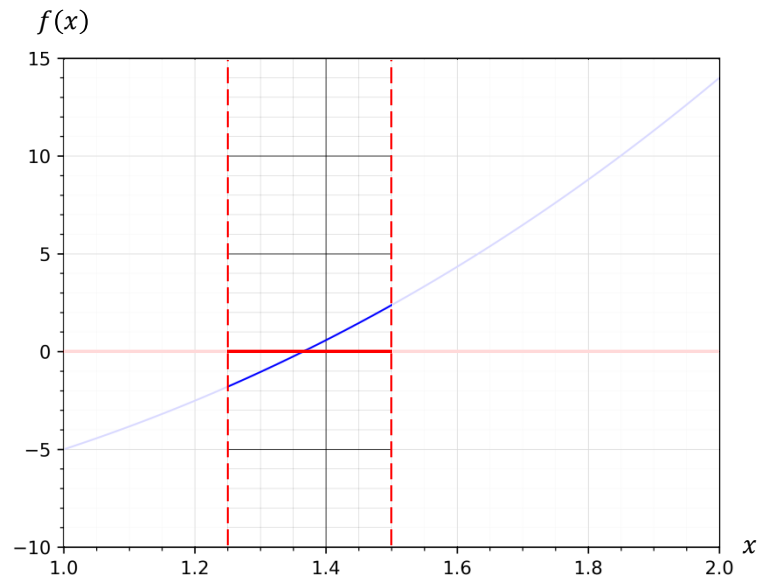
$$a^{(1)} = a^{(0)} = 1, b^{(1)} = p^{(0)} = 1.5 \quad \rightarrow \quad p^{(1)} = \frac{1+1.5}{2} = 1.25$$

$$\varepsilon^{(1)} = \frac{|p^{(1)} - p^{(0)}|}{|p^{(1)}|} = \frac{|1.25 - 1.5|}{|1.25|} = 0.2 > \varepsilon_t \quad \rightarrow \quad \text{continue}$$

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$$f(a^{(1)}) = -5, f(p^{(1)}) = -1.797 \quad \rightarrow \quad f(a^{(1)}) \cdot f(p^{(1)}) > 0$$

$$a^{(2)} = p^{(1)} = 1.25, b^{(2)} = b^{(1)} = 1.5$$



Bisection Method

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$$a^{(0)} = 1, b^{(0)} = 2 \quad \rightarrow \quad p^{(0)} = \frac{1+2}{2} = 1.5$$

$k = 1$

$$f(a^{(0)}) = -5, f(p^{(0)}) = 2.375 \quad \rightarrow \quad f(a^{(0)}) \cdot f(p^{(0)}) < 0$$

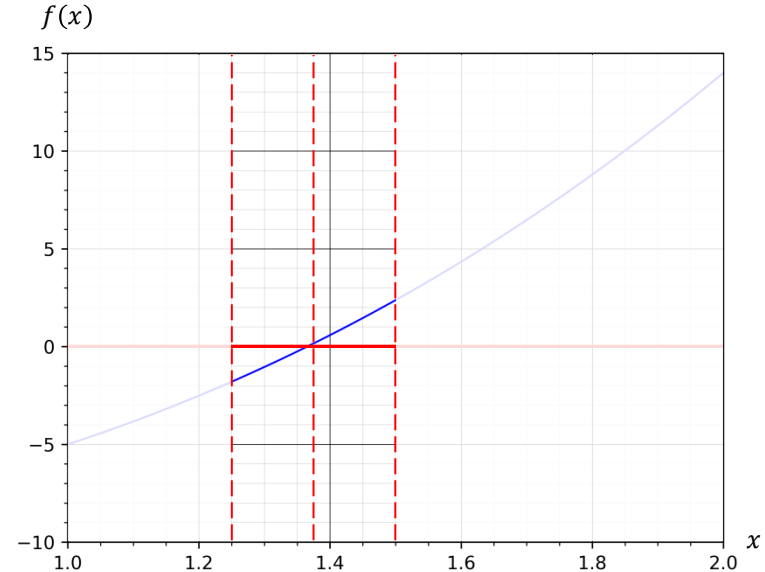
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$$a^{(2)} = p^{(1)} = 1.25, b^{(2)} = b^{(1)} = 1.5 \quad \rightarrow \quad p^{(2)} = \frac{1.25+1.5}{2} = 1.375$$



Bisection Method

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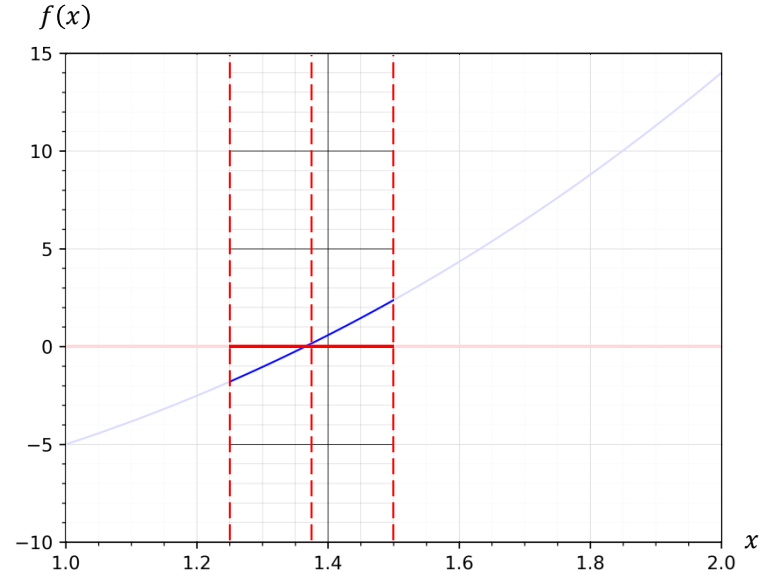
$$\varepsilon^{(1)} = \frac{|p^{(1)} - p^{(0)}|}{|p^{(1)}|} = \frac{|1.25 - 1.5|}{|1.25|} = 0.2 > \varepsilon_t \quad \rightarrow \quad \text{continue}$$

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$$f(a^{(1)}) = -5, f(p^{(1)}) = -1.797 \quad \rightarrow \quad f(a^{(1)}) \cdot f(p^{(1)}) > 0$$

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$$\varepsilon^{(2)} = \frac{|p^{(2)} - p^{(1)}|}{|p^{(2)}|} = \frac{|1.375 - 1.25|}{|1.375|} = 0.090909 > \varepsilon_t$$



Bisection Method

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$k = 1$

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